ACS OMEGA

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Cite This: ACS Omega 2019, 4, 10160−10170

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Elucidation of Calcite Structure of Calcium Carbonate Formation Based on Hydrated Cement Mixed with Graphene Oxide and Reduced Graphene Oxide

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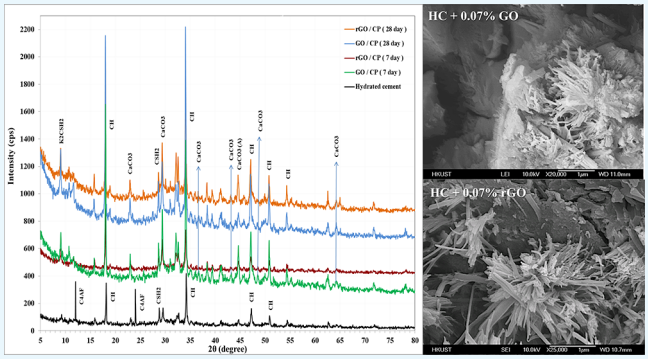
ABSTRACT: In this study, the carbonation of Portland cement by direct chemical interaction with graphene oxide (GO) and reduced graphene oxide (rGO) at 7 and 28 days was examined. During the carbonation reaction, the calcium-bearing phases (calcium hydroxide, calcium silicate hydrate, and ettringite) formed calcium carbonate polymorphs, along with amorphous silica gel, gypsum, and alumina gel. These reaction products were examined using XRD (X-ray diffraction), XPS (energy-dispersive spectrometry), and FTIR (Fourier transform infrared). XRD patterns showed that the intensities of the calcium hydroxide and calcium carbonate peaks in the hydrated cement mixed with GO and/or rGO are higher than the corresponding peaks in the hydrated cement without any additives. The morphology of the reaction products was also characterized by SEM (scanning electron microscopy) measurements, which showed that a needle-like phase of calcium carbonate develops on the hydrated cement. The obtained microstructure parameters enabled the development of a more precise carbonation model.



1. INTRODUCTION

Recently, the increased attention on civil infrastructure enhancement has motivated research on the possible modification of cement-based materials to fulfill the growing requirements for sustainability in the construction field.[1,2] Modification is performed by some selective additives and adjusting the cement composition. Several attempts have been done to enhance the hardness, durability, and tensile properties of the cement by utilizing different kinds of reinforcing materials, from carbon fibers, steel fibers, and polymer fibers[3−11] to nanomaterials,[12,13] such as carbon nanotubes, graphene derivatives, and nanosilica.[14−17] In addition, multiple studies have been committed to investigate the carbonation effects on the reinforced cement durability,[2,18−26] and environmental factors such as temperature, relative humidity, and partial pressure of carbon dioxide have been found to have enormous effects on the carbonation of hydrated cement.[19,20,24−26] Indeed, the carbonation of hydrated cement implies an intricate effect on the microstructure;[27−30] mild CO_2 exposure enhances the compressive strength, while extended exposure might negatively influence the mechanical properties. More importantly, Pavlik[21,22] demonstrated that the reaction of the hydrated cement with atmospheric CO_2 is generally a slow process and is highly dependent on some environmental factors such as the concentration of atmospheric CO_2, temperature, and permeability of the material. Moreover, a few studies have been conducted on the combination of graphene oxide (GO) with cement-based materials,[31−34] but they were essentially focused on improving microstructural and mechanical properties, without realizing the possibility of the existence of a direct chemical interaction between GO and hydrated cement. Li et al.[35] showed that the addition of 0.04 wt % GO enhanced the tensile strength by 67% in comparison with the plain cement paste. Lv et al.[37] indicated that the addition of GO at ratios of 0.03, 0.05, and 0.07% increases the compressive strength values of the GO/cement composites from 150 to 177 MPa. With the 0.05% GO ratio, the compressive and flexural strengths were 176.64 and 31.6 MPa, respectively. Lu et al.[36] demonstrated that the addition of 0.05% GO to the cement paste increased the compressive and flexural strengths to 11.1 and 16.2%, respectively. Until now, there have been no reports on the direct interaction mechanism of GO with hydrated cement-based materials, although we previously reported that GO and/or reduced GO (rGO) in their direct chemical interaction with either $Ca(OH)_2$ or CaO could be used to synthesize pure

Received: January 6, 2019
Accepted: May 29, 2019
Published: June 12, 2019

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DOI: 10.1021/acsomega.9b00042
ACS Omega 2019, 4, 10160−10170

polymorph $CaCO_3$ crystals with consistent morphology and size.[38,41−45] This is related to the oxygen functional groups that decomposed on the surfaces of GO or rGO in specific alkaline media releasing CO, CO_2, and water.[38,41−46] The objective of the current research was to elucidate the mechanism of the chemical interaction of GO or rGO with hydrated cement-based materials, which is related to the oxygen functional groups that decomposed on the surfaces of GO or rGO in specific alkaline media to react with calcium-bearing phases in cement paste (CP), such as calcium hydroxide (CH), calcium silicate hydrate (C-S-H), and ettringite (AFt), which contributed substantially to $CaCO_3$ polymorph formation, and showed that the formed $CaCO_3$ amount far exceeded that which could be obtained from complete CH dissolution. On the other hand, the chemical interaction of the cement paste (CP) with GO or rGO could be utilized to achieve specific economic and environmental benefits. This can be viewed as a promising opportunity to take advantage of this chemical reaction.

2. RESULTS AND DISCUSSION

2.1. Mechanism of GO and rGO Reaction with Hydrated Cement. Portland cement is a mixture of heterogeneous compounds in an anhydrous state. This mixture originally consists of alite (C_3S, Ca_3SiO_5), belite (C_2S, Ca_2SiO_4), tricalcium aluminate (C_3A, $Ca_3Al_2O_6$), and tetracalcium aluminoferrite (C_4AF, $Ca_4(Al,Fe)_2O_{10}$) as well as limited quantities of other impurities, such as magnesium sulfate ($MgSO_4$), sodium sulfate (Na_2SO_4), potassium sulfate (K_2SO_4), and gypsum ($CaSO_4{\cdot}2H_2O$).[1,2,47−49] During normal hydration, Portland cement will undergo a complex hydration reaction to form CH ($Ca(OH)_2$), C-S-H gel ($3CaO{\cdot}2SiO_2{\cdot}4H_2O$), and ettringite (AFt, $CaAl_2(SO_4)_3(OH)_{12}{\cdot}26H_2O$ and AFm, $Ca_4Al_2(OH)_2{\cdot}SO_4{\cdot}6H_2O$).[1,44,45] In this regard, the direct chemical interactions of GO or rGO (with pH 3.01) with hydrated cement have a neutralizing effect on the highly alkaline medium of CP with a pH greater than 12.5 in the pore solution, inducing the acidic nature of the oxygen functional groups on the graphene layer surfaces in GO or rGO, such as carboxyl, hydroxyl, and epoxy.[18,21−23,38−46]

2.1.1. CH-Phase Carbonation. In a lower alkaline environment, CH (portlandite, $Ca(OH)_2$) is the first component to start dissolution,[24−26] inducing the process of the oxygen functional group decomposition at the surfaces of GO or rGO to release CO_2, CO, and H_2O, which appears to be responsible for calcium carbonate ($CaCO_3$) polymorph formation[18,24,38−46,52−54] according to eq 1:

$$nCa(OH)_2 + nGO + nH_2O \rightarrow nCaCO_{3(s)} + (n+1)H_2O \quad (1)$$

We believe that the reaction of GO or rGO with hydrated cement is governed by the total amount of the oxygen functional groups located on the surfaces of GO and/or rGO. The determination of oxygen groups on the surfaces of various carbon materials was analyzed by both the Boehm titration and the thermal-programmed desorption (TPD) methods.[41−45,65] TPD gives quantitative information about the total number of surface oxygen groups, which break down at various temperatures to release CO, CO_2, and H_2O, even though the Boehm titration method provides both qualitative and quantitative information solely regarding acidic and basic groups (which include phenols, lactone groups, and carboxylic acids). The acidic nature of the oxygen functional groups could be equalized by bases of various strengths such as $NaHCO_3$, Na_2CO_3, and NaOH.

2.1.2. C-S-H-Phase Carbonation. The results strongly indicate that the direct reaction of C-S-H with GO or rGO results in the formation of amorphous silica gel and various $CaCO_3$ polymorphs according to eq 2:

$$nCaO{\cdot}SiO{\cdot}mH_2O + nGO + nH_2O \rightarrow nCaCO_{3(s)} + nSiO_2(H_2O)_m + (n-m)H_2O \quad (2)$$

2.1.2.1. C-S-H Gel. As stated in our previous work,[38] the direct reaction of GO or rGO with $Ca(OH)_2$ or CaO results in $CaCO_3$ formation. Thus, the rate and degree of the carbonation of the C-S-H gel depend greatly on the amounts of CO_2 and CO released during decomposition, which are associated to the total amount of the oxygen functional groups located on the surfaces of GO or rGO.[38−46] It is well established that carboxyl and lactone groups are responsible for CO_2 production, while aldehyde, carbonyl, and phenols are responsible for the production of CO.[40−45] Taking into account weathering carbonation upon extended exposure to atmospheric CO_2 in the process, C-S-H is believed to be more decalcified and to lose its binding ability accordingly.[24−26,53,54] Carbonation of the C-S-H gel generally results in calcite, aragonite, and vaterite formation.[26] Aragonite and vaterite formation are apparently related to the existence of highly decalcified C-S-H and consequently to high CO_2 (GO or rGO decomposition product) concentrations.[24−26,38−46,52−54] Moreover, we expect that the direct chemical interactions of GO or rGO with hydrated cement-based materials do not necessarily result in full carbonation on the surface of the cement paste but rather that a completely carbonated zone can exist in the core of the paste.

2.1.3. Ettringite-Phase Carbonation. Our results also emphasize the effective chemical interactions of GO or rGO with other calcium-bearing phases. Apart from the CH and C-S-H phases, ettringite and aluminates are other cement phases that decompose upon reaction with GO or rGO, resulting in gypsum ($CaSO_4{\cdot}2H_2O$) and alumina gel ($Al_2O_3{\cdot}mH_2O$) formation according to eq 3:

$$nCaO{\cdot}Al_2O_3{\cdot}nCaSO_4{\cdot}mH_2O + nGO + nH_2O \rightarrow nCaCO_{3(s)} + n(CaSO_4.\,2H_2O) + Al_2O_3{\cdot}mH_2O + (n-m)H_2O \quad (3)$$

The carbonation of ettringite may lead to vaterite crystal formation.[55−57] Carbonation of all calcium-bearing phases, such as CH, C-S-H, and ettringite, contributes substantially to $CaCO_3$ polymorph formation, and it is frequently mentioned that the formed $CaCO_3$ amount far exceeded that which could be obtained from the entire CH dissolution.[24,52−58]

Moreover, during carbonation, both GO and rGO eventually decompose to form CO_2, CO, and H_2O, which completely react with hydrated cement (calcium-bearing phases, CH, C-S-H, and ettringite) to form $CaCO_3$. The Boehm titration method reported that the acidity of the oxygen functional groups on the carbon materials can be equalized by bases of various strengths. The oxygen functional groups decompose at various temperatures to release CO, CO_2, and water. During decomposition, the released CO and CO_2 associated to the total amount of the oxygen functional groups located on the

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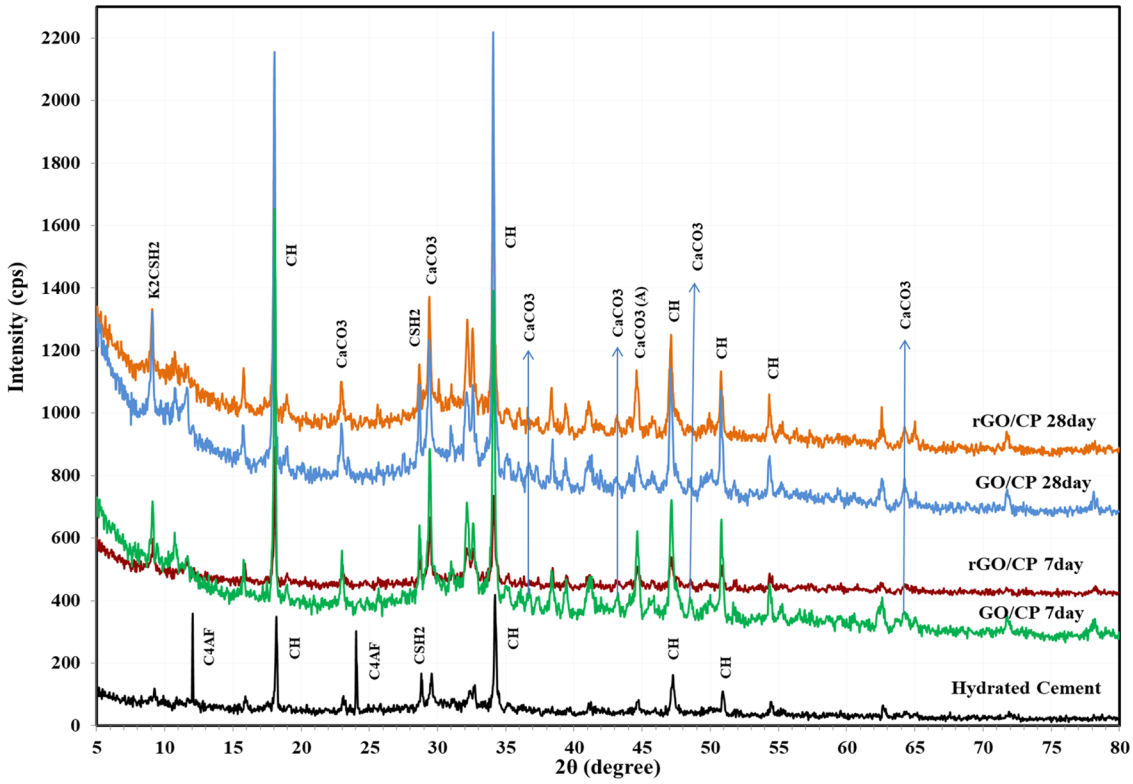

Figure 1. XRD patterns of hydrated cement and hydrated cement mixed with 0.07% of GO and 0.07% of rGO at 7 and 28 days.

surfaces of GO or rGO. It is well established that carboxyl and lactone groups are responsible for CO_2 production, while aldehyde, carbonyl, and phenols are responsible for the production of CO.[41−45,65] Calcium carbonate ($CaCO_3$) formation was detected using Fourier transform infrared (FTIR) spectra, X-ray diffraction (XRD) patterns, etc., as will be discussed later. Therefore, this mechanism emphasizes the chemical interaction between hydrated cement and GO or rGO, resulting in the formation of $CaCO_3$ (calcite polymorph), regardless of the variation between GO and rGO, specifically in terms of their physical properties, and the surface morphologies of GO and rGO.[38]

Overall, the carbonation of hydrated cement (formation of $CaCO_3$ with silica gel) can be useful for cement-based materials, as it has previously been reported to enhance the compressive and flexural strengths of cement and reduce the atmospheric carbonation of hydrated cementitious matrices.[24,31−34] Conversely, the prolonged carbonation of hydrated cement, such as CH and the C-S-H phase, may lead to corrosion of reinforcing steel bars due to a pH decrease to a certain level (less than 9.7), which may cause serious structural damage.

2.2. XRD Characterization. XRD characterization was implemented to identify the crystalline phases of the solid products. A representative set of diffraction data for the four hydrated cements is shown in Figure 1.

A comparison of the peak intensities between hydrated cement and GO and/or rGO provides some insight into their relative compositions. Evidently, the acidic nature of the oxygen functional groups on the graphene-derivative layers, such as epoxy, hydroxyl, and carboxyl,[18,21−23,38−46] on the surfaces of GO or rGO (with pH 3.01) has a neutralizing effect on the highly alkaline medium of the hydrated cement with a pH greater than 12.5 in the pore solution. Therefore, the intensities of the CH and $CaCO_3$ peaks in hydrated cement mixed with GO and/or rGO are higher than the corresponding peaks in the hydrated cement without any additives.

The XRD patterns also show the $CaCO_3$ peak intensity, which increases with GO and/or rGO addition due to the increase in CH, which is available for carbonation as a result of the presence of the oxygen functional groups that decompose on the surfaces of GO or rGO. This process releases CO_2, CO, and H_2O,[38−46] which can be precipitated in one or a mix of various polymorphs of $CaCO_3$ (calcite, aragonite, and vaterite). Calcite appears to be the primary crystalline $CaCO_3$ polymorph, with diffraction peaks at 2θ = 23°, 29.3°, 36°, 43°, 48.4°, and 64.3°, which are attributed to calcite. Characteristic peaks of vaterite and aragonite at 2θ = 25° and 45°, respectively, are evident, but they are relatively low compared with that of calcite. However, the XRD patterns show sharp calcite reflections, confirming the effective formation of calcite.[20,24−26,48,51,59] Further, a comparison of the XRD patterns for the $CaCO_3$ particles acquired using GO and/or rGO reveals that the XRD peaks for the hydrated cement mixed with GO and/or rGO are sharper than those of the hydrated cement without any additives.

Typical energy-dispersive spectrometry (XPS) spectra of the chemical interaction of GO or rGO with hydrated cement at 7 and 28 days are shown in Figure 2. The original and fitting curves of the $Ca(OH)_2$ and $CaCO_3$ phases match remarkably well.

Indeed, a GO or rGO reaction with calcium-bearing phases (CH and C-S-H) in hydrated cement can cause chemical−mechanical changes in the microstructure.[24] Due to the acidic nature of the oxygen functional groups on the surfaces of GO or rGO, they will decompose in the highly alkaline medium releasing CO, CO_2, and H_2O, leading to faster carbonation,

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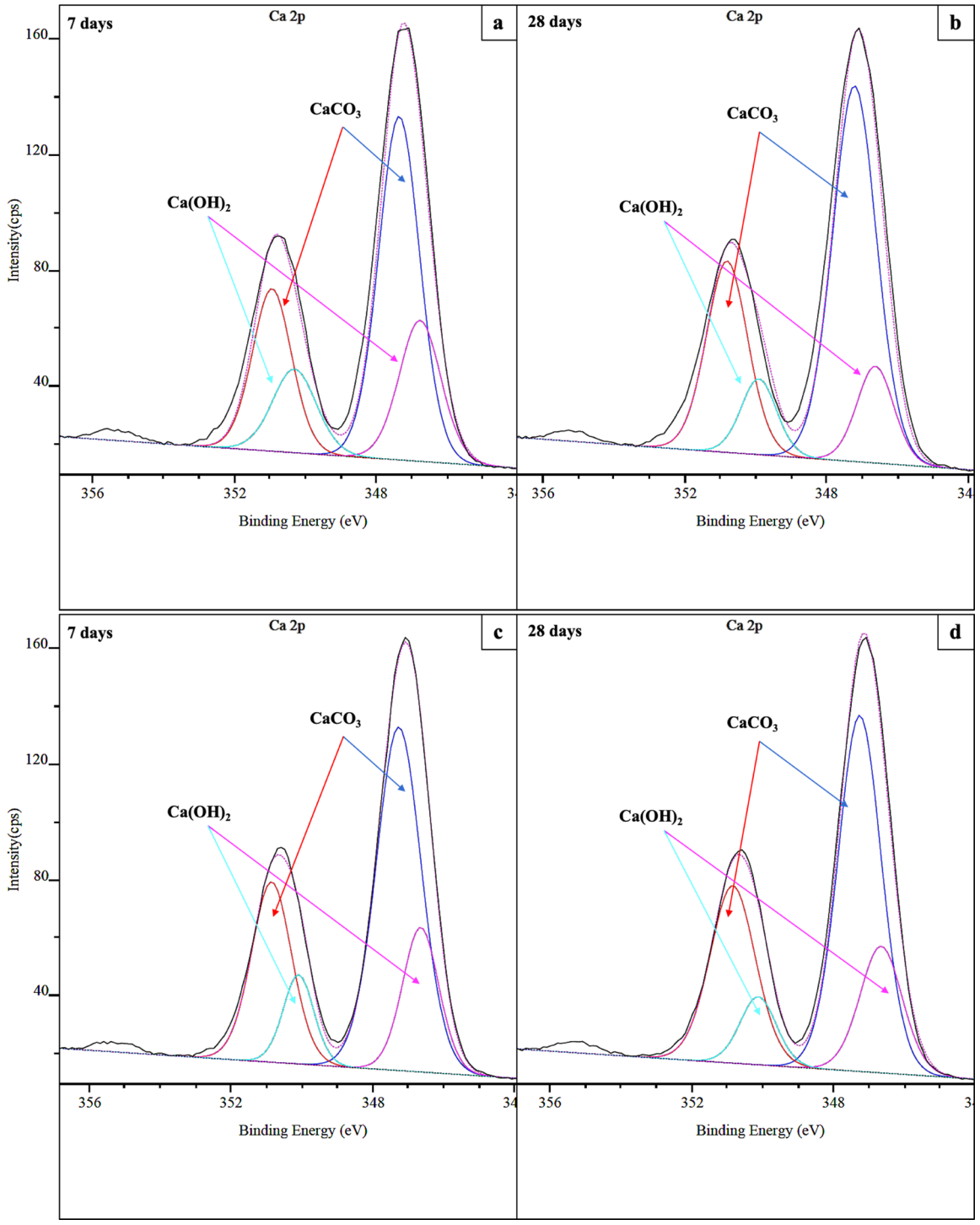

Figure 2. XPS spectra of Ca 2p for hydrated cement: (a, b) with 0.07% of GO and (c, d) with 0.07% of rGO at 7 and 28 days.

more $Ca(OH)_2$ consumption, and consequently more $CaCO_3$ generation.[25,29,38,60]

2.3. FTIR Spectrum Analysis. The FTIR spectra of hydrated cement and hydrated cement mixed with GO and/or rGO at 7 and 28 days are shown in Figures 3 and 4. The positions of all of the infrared active modes of calcite found in hydrated cement mixed with GO and/or rGO precisely match those of the calcite reference bands.[38,50,64] It can be seen from the infrared spectra (Figures 3 and 4) of hydrated cement mixed with GO and/or rGO that asymmetric stretching (v3) gives rise to a very sharp and broad infrared absorption at 1425 cm^{-1}. Out-of-plane and in-plane bending (v2 and v4) correspond to the fundamental bands of calcite at 872 and 713 cm^{-1}, respectively.[19,38] The clear presence of these absorption bands confirms the formation of the calcite phase of $CaCO_3$ particles when either GO or rGO is mixed with hydrated cement.[38] It should be noted that no residual peaks of GO or rGO are observable, confirming our predictions about the eventual decomposition to form CO_2, CO, and H_2O.[38−46] In addition, it has been reported that thermal-programmed desorption (TPD) was used to characterize the oxygenated functional groups on the surfaces of the carbon materials,

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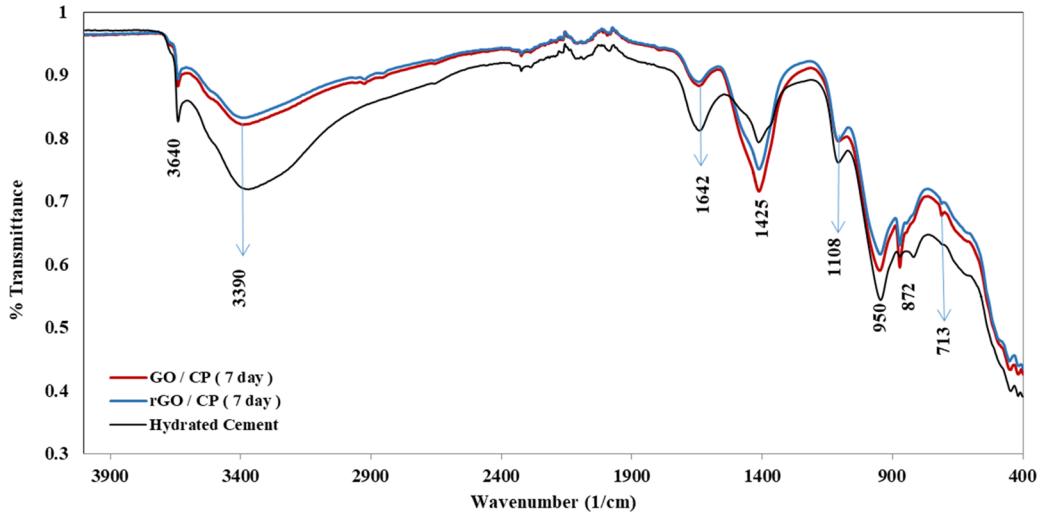

Figure 3. FTIR spectra of hydrated cement and hydrated cement mixed with 0.07% of GO and 0.07% of rGO at 7 days.

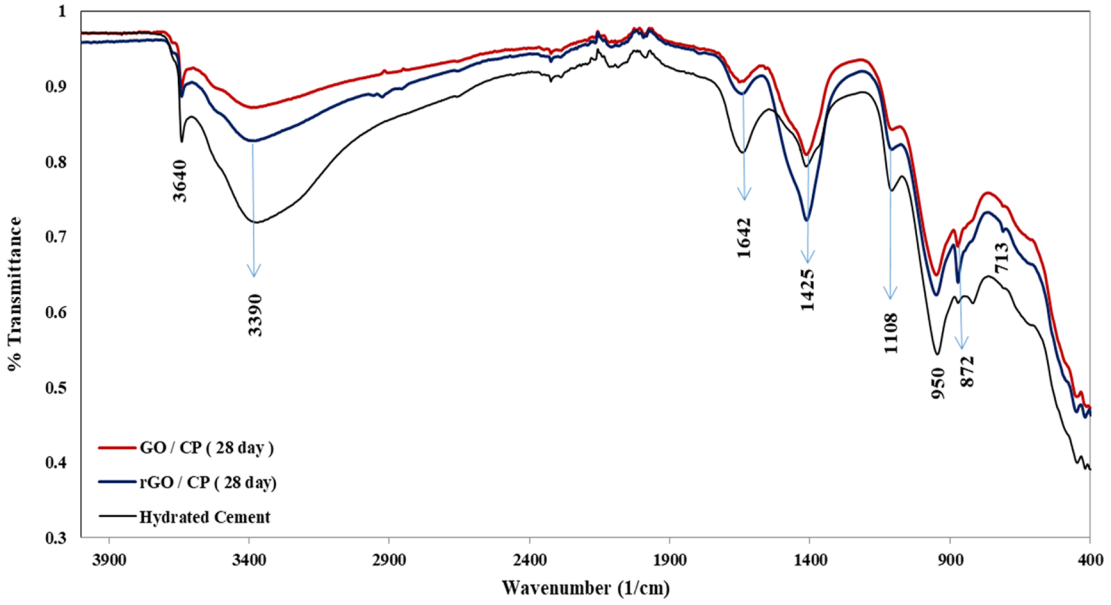

Figure 4. FTIR spectra of hydrated cement and hydrated cement mixed with 0.07% of GO and 0.07% of rGO at 28 days.

which break down at various temperatures releasing CO, CO_2, and H_2O.[65] This analysis also confirms the XRD and XPS analysis results presented in Figure 1. In addition, it is crucial to note that more intense/sharper characteristic peaks are observable in the FTIR spectra for the calcite particles acquired using either GO or rGO with hydrated cement than in the spectra for the hydrated cement without any additives. The peaks at 450 cm^{-1} falls in the range of 400−500 cm^{-1}, which corresponds to the Si−O bending vibration (v2) in SiO_4^{-2}, according to refs 19 and 50. Furthermore, both spectra show peaks at 950 cm^{-1}, corresponding to asymmetric Si−O stretching vibrations, which indicates limited polymerization in C-S-H.[19,50,61]

Moreover, the four spectra show weak bands at 1108 cm^{-1}, falling in the range of 1100−1165 cm^{-1}, which corresponds to SO_4^{-2} vibration (v3) in sulfates.[50,62,63] This region has a range of peaks that may overlap due to the polymerization of SiO_4^{-2} and the corresponding vibration (v3).[48,50] The water band peaks of the four spectra, which are located at 1642 cm^{-1}, indicate the H−O−H bending vibration (v2) of the adsorbed water molecule.[19,50]

Several additional weak bands between 3000 and 2000 cm^{-1} are overtone modes. Meanwhile, the peaks at 3390 cm^{-1}, which fall in the range of 3100−3400 cm^{-1}, correspond to the O−H stretching vibrations (v1 and v3) in the water molecules.[19,48] The peaks at 3640 cm^{-1} (Figures 3 and 4) correspond to CH, which is formed as silicate phases in the hydrated cement. It is clear that the intensities of the characteristic peaks of CH decrease markedly when either GO and/or rGO is mixed with hydrated cement, compared to their intensities in the spectra of the hydrated cement without any additives. This difference is mainly due to the reaction of CH with liberated CO_2 from the decomposition of the oxygen functional groups of GO or rGO.[19,38−46,48,50,62,63,65] These findings also confirm the XRD and XPS analysis results discussed previously and prove that GO and/or rGO have high reactivity with hydrated cement, leading to nano-calcium carbonate generation.[38] Nano-calcium carbonate can block the

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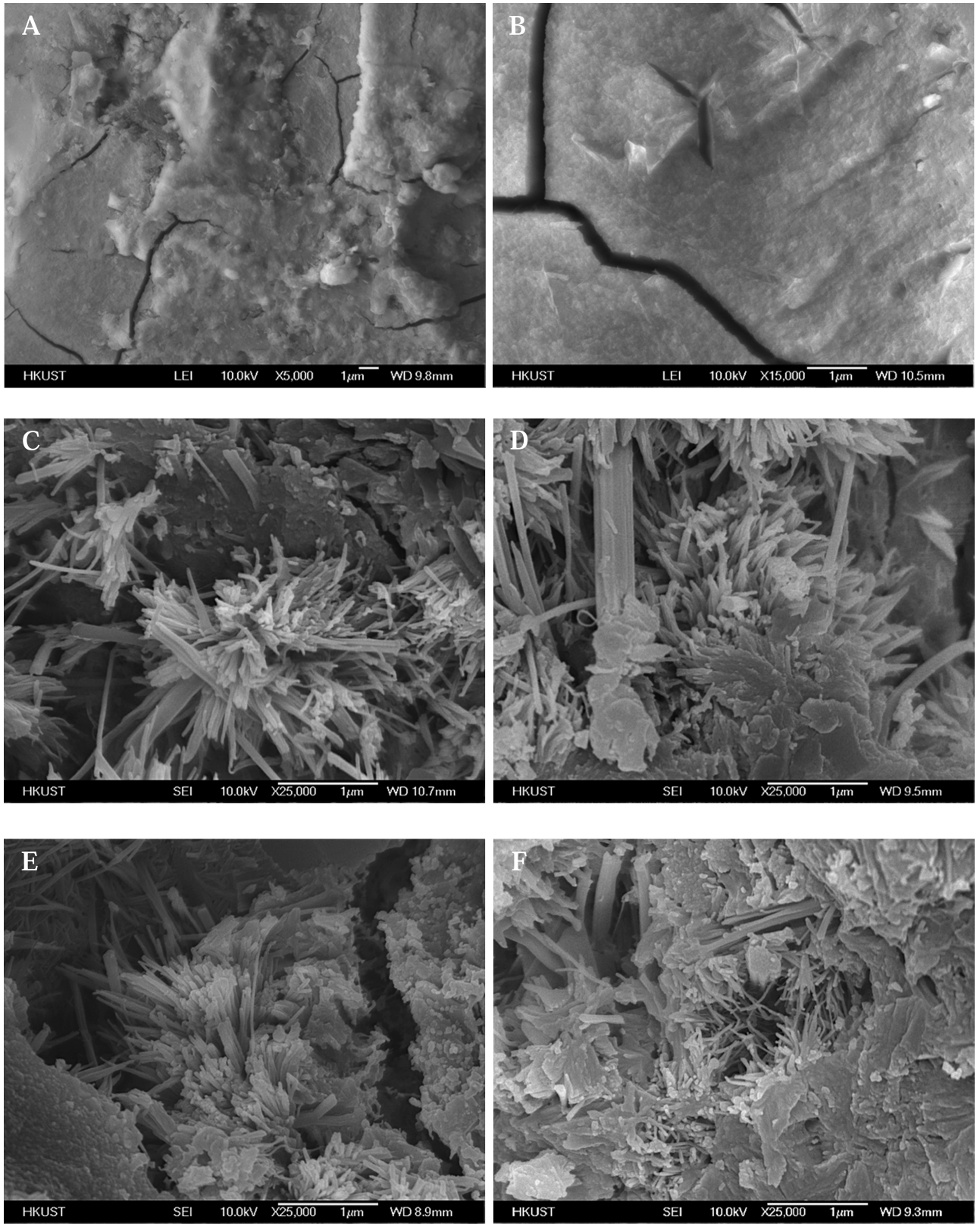

Figure 5. SEM images of the microstructure of hydrated cement: (A, B) without GO and/or rGO, (C, D) with 0.07% of GO, and (E, F) with 0.07% of rGO at 7 and 28 days.

pores in the hardened cement, which may or may not improve the mechanical properties.

2.4. SEM. The microstructures of hydrated cement mixed with 0.07% of GO and/or rGO were examined using scanning electron microscopy (SEM). The SEM results presented in Figures 5 and 6 depict the microstructures of the hydrated cement mixed with GO and/or rGO at 7 and 28 days. Figure 5A,B shows the microstructure of the hydrated cement without

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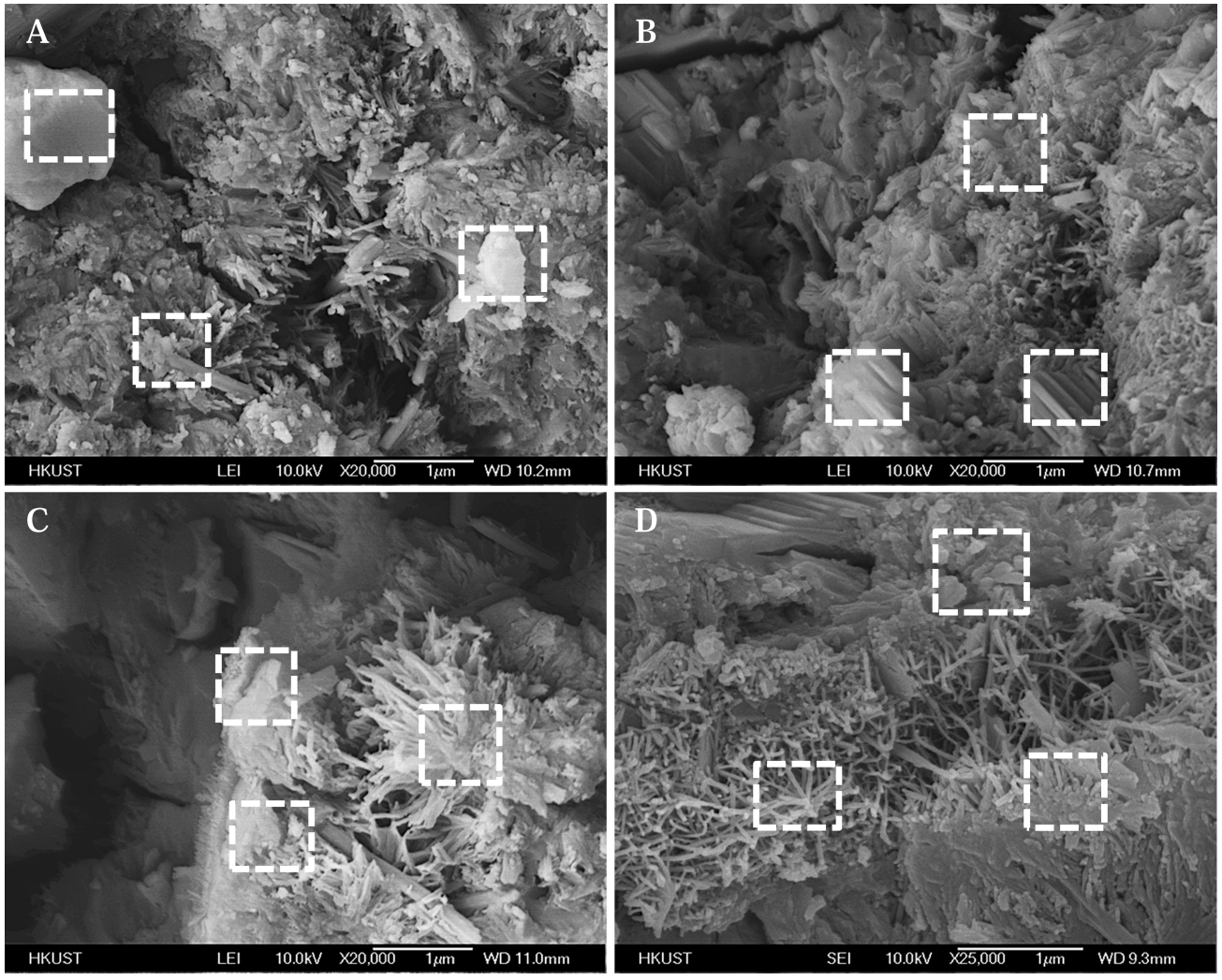

Figure 6. SEM images of the microstructure of hydrated cement at different locations and magnifications: (A, B) with 0.07% of GO and (C, D) with 0.07% of rGO at 7 and 28 days. (The areas with white markings in the SEM images are selected areas for EDX detection.)

any additives, proving that the microstructure is an amorphous solid with a large amount of microcracks, porosity, and pore size distribution. When compared with Figure 5C–F, the results prove that the hydrated cement composites mixed with a 0.07% dosage of GO and /or rGO have a certain amount of needle-like crystal products and form compacted microstructures, which strongly prove that, due to the acidic nature of the oxygen functional groups located on the surfaces of GO or rGO, they will decompose in a highly alkaline medium releasing CO, CO_2, and H_2O,[38,41−45] which accelerate the dissolution rate and the hydration of cement at the first few minutes of the starting reaction. The acceleration phase of $CaCO_3$ formation starts due to the direct carbonation reaction of cement hydrates with GO and/or rGO at different times (Figure 5C–F). Moreover, the results indicate that the direct reaction of GO and/or rGO with hydrated cement produces a certain amount of $CaCO_3$ crystals that participate in forming the cross-linking and interweaving microstructure, which leads to enhancing the mechanical behavior of the hydrated cement paste. Figure 6A–D shows different portions and magnifications of a relatively higher growth of the calcite phase of $CaCO_3$ in the hydrated cement mixed with GO and/or rGO at 7 and 28 days. Simultaneously, a needle-like phase develops on the cement particles. This needle-like phase has been linked to the calcite structure in earlier studies.[38,48,50] The distribution of $CaCO_3$ crystals in hydrated cement composites could be identified by inspecting the $CaCO_3$ crystals in a whole SEM image using energy-dispersive X-ray spectroscopy (EDX). The EDX-inspected areas are marked as white boxes in Figure 6A–D. The EDX test results indicate that the needle-like crystals of $CaCO_3$ is uniformly distributed within the whole inspected area (Figure 7). These results also suggest that the needle-like phase formed continuously during hydration and grew even more after 28 days of hydration. As a matter of fact, the various morphologies of $CaCO_3$ particles resulted from the variations in the concentration and type of the oxygen functional groups on the surfaces of GO and rGO, which serve as active sites and play an essential role in such reactions.[38−45] However, both GO and rGO eventually decompose to form CO_2, CO, and H_2O in a highly alkaline medium, which completely react with hydrated cement (calcium-bearing phases, CH, C-S-H, and ettringite) to form needle-like crystals of $CaCO_3$. These findings also confirm the XRD, XPS, and FTIR analysis results and prove that GO and/or rGO have high reactivity with hydrated cement.

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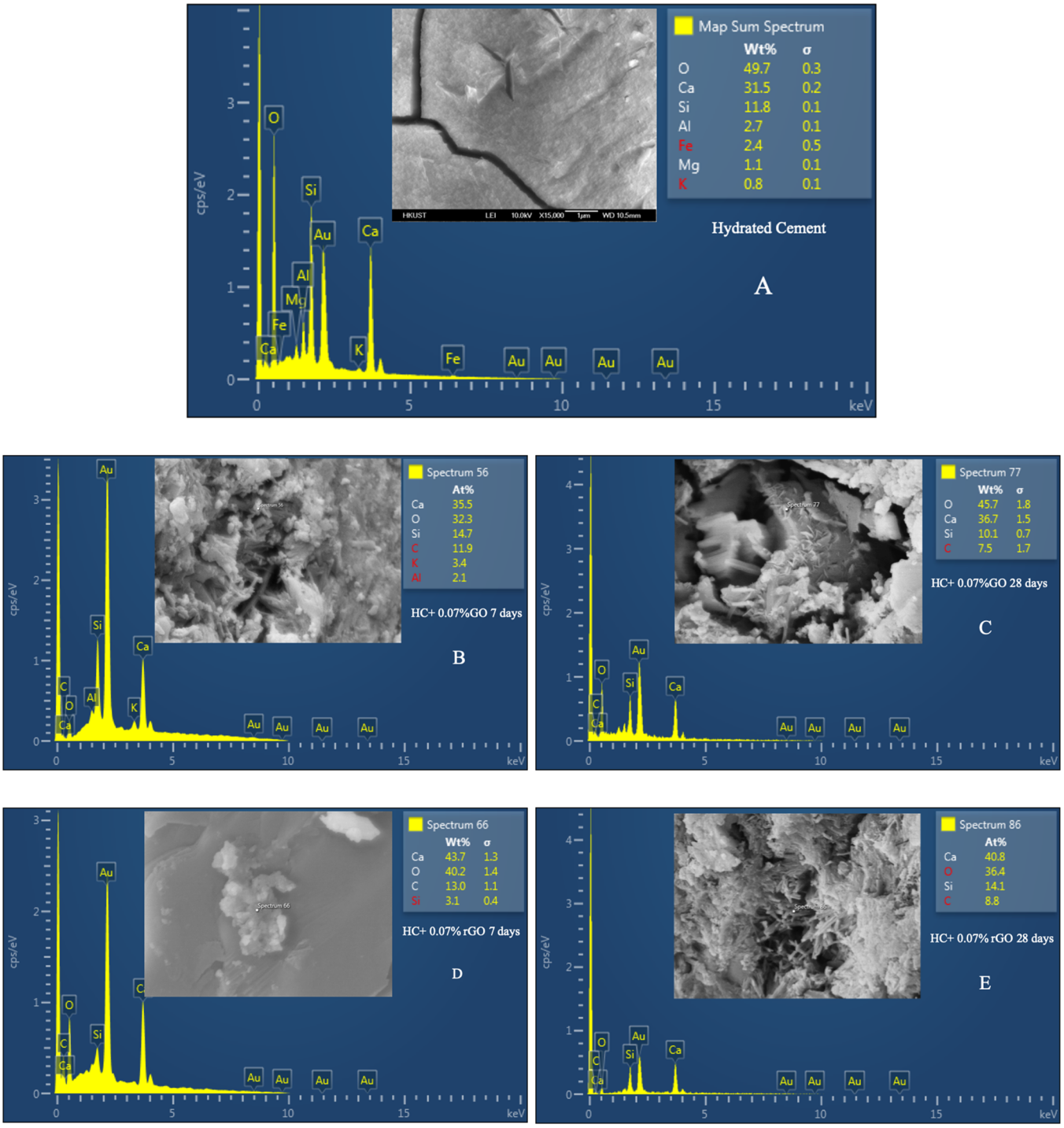

Figure 7. EDX analysis of $CaCO_3$ particles obtained: (A) hydrated cement, (B, C) hydrated cement mixed with 0.07% of GO at 7 and 28 days, and (D, E) hydrated cement mixed with 0.07% of rGO at 7 and 28 days

3. CONCLUSIONS

The development of GO and rGO cementation actions is a significant area of interest in the construction field. The impressive ability of GO to release CO_2, CO, and H_2O through the decomposition of the oxygen functional groups in certain alkaline media, which appears to be behind the carbonation of Portland cement, is a phenomenon that could be employed to elucidate the mechanism of this chemical interaction. Toward this end, this study was focused on investigating the critical reaction mechanism using GO, rGO, and Portland cement and on how these materials may influence CH, C-S-H, and ettringite carbonation. The rate and degree of CH and C-S-H gel carbonation depend fully on the quantities of CO_2 and CO released during decomposition, which are associated to the total amount of the oxygen functional groups located on the surfaces of GO or rGO. These processes result in an obvious pH reduction to a certain level (less than 9.7). The ettringite and aluminate phases decompose upon reaction with GO or rGO, resulting in gypsum and alumina gel formation. XRD, XPS, and FTIR were used to confirm the $CaCO_3$ polymorph formation. The SEM

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Table 1. Chemical Composition of Cement

component	CaO	SiO_2	SO_4	Al_2O_3	Fe_2O_3	MgO	K_2O	TiO_2
content (%)	65.40	19.47	5.71	3.86	3.00	1.58	0.49	0.26

images also corroborated the needle-like phase attributed to the calcite polymorph. Principally, the current study has evidently established the usefulness of GO and/or rGO with hydrated cement for obtaining $CaCO_3$, which could be beneficial for the cement industry, as it has previously been reported to increase the compressive and flexural strengths of CP and reduce the atmospheric carbonation of hydrated cementitious matrices.

4. EXPERIMENTAL SECTION

4.1. Preparation of Cement−GO. GO was synthesized from graphite powder (200 mesh, Alfa-Aesar) utilizing the Hummers' method.[38,40] Ordinary Portland cement (type 52.5, Green Island, Hong Kong) was utilized in this work, and the chemical composition collected by X-ray fluorescence is listed in Table 1. Table 2 details the mixture ratios of the GO and/or rGO with modified cement paste. The mixture was prepared first by ultrasonication of the aqueous solution of GO for 15 min, which was then slowly mixed with cement (at a ratio of 0.4 water to cement by weight) at the speed of a high rate for 8 min before casting into molds and covering with plastic sheets. The samples were cured in ambient conditions for 24 h, demolded, and placed into a curing room (25 °C/RH 95%) until testing.

Table 2. Mixed Proportions of GO- and rGO-Modified Cement Paste

sample	cement (g)	H_2O (g)	GO (mL)	rGO (g)	dosage (wt %)
hydrated cement	400	140.0			0.00
cement−GO	400	70.0	70.0		0.07
cement−rGO	400	70.0		70.0	0.07

4.2. Preparation of Cement−rGO. rGO was prepared following the same method that was reported previously.[38] Table 2 details the mixture ratios of the GO and/or rGO with modified cement paste. The mixture was prepared first by ultrasonication of the aqueous solution of rGO for 15 min, which was then slowly mixed with cement (w/c = 0.4 by weight) at the speed of a high rate for 8 min before casting into molds and covering with plastic sheets. The samples were cured in ambient conditions for 24 h, demolded, and placed into a curing room (25 °C/RH 95%) until testing. For both GO or rGO with hydrated cement, the samples were aged for 7 and 28 days and subjected to hydration stoppage by immersing in acetone and then drying at 50 °C for 1 h. They were then kept in an air-tight container to prevent any air exposure until observations were performed.

4.3. Material Characterization. The produced samples were characterized by XRD patterns, which were collected by a PW1830 (Philips) system at 2 kW, using a Cu anode and a graphite monochromatic system. XPS measurements were performed by a Kratos Axis Ultra DLD Multitechnique using a monochromatic Al Kα X-ray source. The FTIR spectra were acquired using a Vertex 70 Hyperion 1000 (Bruker) spectrometer in the range of 400−4000 cm^{-1} averaged over 32 scans. The SEM images were carried out in a JEOLJSM-6700F scanning electron microscope equipped with an energy-dispersive X-ray analyzer.

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Notes

The authors declare no competing financial interest.

ACKNOWLEDGMENTS

This study was supported financially by the University of Macau under grant no. SRG2016-00093-IAPME and the Ministry of Science and Technology of China under grant no. 2015CB655100. The authors would also like to acknowledge the assistance from the Materials Characterizations and Preparation Facilities (MCPF) at the Hong Kong University of Science and Technology. The authors acknowledge Zainab A. Yaseen for her help in proofreading the manuscript.

REFERENCES

(1) Mehta, P.; Monteiro, P. *Concrete Microstructure, Properties And Materials*; 3rd ed.; McGraw-Hill: New York, 2006.

(2) Butler, W. B. *Guide To Durable Concrete*; ACI, 2001.

(3) Yuan, Y.; Ji, Y. Modeling Corroded Section Configuration Of Steel Bar In Concrete Structure. *Constr. Build. Mater.* **2009**, *23*, 2461−2466.

(4) Watanabe, K.; Kimura, T.; Niwa, J. Synergetic Effect Of Steel Fibers And Shear-Reinforcing Bars On The Shear-Resistance Mechanisms Of RC Linear Members. *Constr. Build. Mater.* **2010**, *24*, 2369−2375.

(5) Hossain, Z.; Awal, A. S. M. A. Flexural Response Of Hybrid Carbon Fiber Thin Cement Composites. *Constr. Build. Mater.* **2011**, *25*, 670−677.

(6) Ali, M.; Liu, A.; Sou, H.; Chouw, N. Mechanical And Dynamic Properties Of Coconut Fibre Reinforced Concrete. *Constr. Build. Mater.* **2012**, *30*, 814−825.

(7) Reis, J. M. L. Fracture And Flexural Characterization Of Natural Fiber-Reinforced Polymer Concrete. *Constr. Build. Mater.* **2006**, *20*, 673−678.

(8) Soroushian, P.; Tlili, A.; Alhozaimy, A.; Khan, A. Development And Characterization Of Hybrid Polyethylene-Fibre-Reinforced Cement Composites. *Constr. Build. Mater.* **1993**, *7*, 221−229.

(9) Soroushian, P.; Won, J.-P.; Hassan, M. Durability Characteristics Of CO_2-Cured Cellulose Fiber Reinforced Cement Composites. *Constr. Build. Mater.* **2012**, *34*, 44−53.

(10) Su, H.; Xu, J. Dynamic Compressive Behavior Of Ceramic Fiber Reinforced Concrete Under Impact Load. *Constr. Build. Mater.* **2013**, *45*, 306−313.

(11) Morova, N. Investigation Of Usability Of Basalt Fibers In Hot Mix Asphalt Concrete. *Constr. Build. Mater.* **2013**, *47*, 175−180.

(12) Yeşilmen, S.; al-Najjar, Y.; Balav, M. H.; Şahmaran, M.; Yıldırım, G.; Lachemi, M. Nano-Modification To Improve The Ductility Of Cementitious Composites. *Cem. Concr. Res.* **2015**, *76*, 170−179.

(13) Sanchez, F.; Sobolev, K. Nanotechnology In Concrete − A Review. *Constr. Build. Mater.* **2010**, *24*, 2060−2071.

(14) Koohestani, B.; Belem, T.; Koubaa, A.; Bussière, B. Experimental Investigation Into The Compressive Strength Develop-

Downloaded from pubs.acs.org/acsodf/article-pdf/4/6/10160/6279164/ao9b00042.pdf by guest on 18 August 2026

DOI: 10.1021/acsomega.9b00042
ACS Omega 2019, 4, 10160−10170

ment Of Cemented Paste Backfill Containing Nano-Silica. *Cem. Concr. Compos.* **2016**, *72*, 180−189.
(15) Du, H.; Du, S.; Liu, X. Effect Of Nano-Silica On The Mechanical And Transport Properties Of Lightweight Concrete. *Constr. Build. Mater.* **2015**, *82*, 114−122.
(16) Konsta-Gdoutos, M. S.; Metaxa, Z. S.; Shah, S. P. Multi-Scale Mechanical And Fracture Characteristics And Early-Age Strain Capacity Of High Performance Carbon Nanotube/Cement Nanocomposites. *Cem. Concr. Compos.* **2010**, *32*, 110−115.
(17) Li, G. Y.; Wang, P. M.; Zhao, X. Mechanical Behavior And Microstructure Of Cement Composites Incorporating Surface-Treated Multi-Walled Carbon Nanotubes. *Carbon* **2005**, *43*, 1239−1245.
(18) Allahverdi, A.; Škvára, F. Acidic Corrosion of Hydrated Cement Based Materials Part 1.- Mechansim of the Phenomenon. *Ceramics − Silikáty.* **2000**, *44*, 114−120.
(19) Ylmén, R.; Jäglid, U. Carbonation Of Portland Cement Studied By Diffuse Reflection Fourier Transform Infrared Spectroscopy. *Int. J. Concr. Struct. Mater.* **2013**, *7*, 119−125.
(20) Nakano, K.; Mito, S.; Xue, Z.; Ohbuchi, A. Observation Of Cement/Sandstone Interface After Reaction With Supercritical CO2 Using SEM-EDS, M-XRD, And M-Raman Spectroscopy. *e-J. Surf. Sci. Nanotechnol.* **2016**, *14*, 198−203.
(21) Pavlík, V. Corrosion Of Hardened Cement Paste By Acetic And Nitric Acids Part II: Formation And Chemical Composition Of The Corrosion Products Layer. *Cem. Concr. Res.* **1994**, *24*, 1495−1508.
(22) Pavlík, V. Corrosion Of Hardened Cement Paste By Acetic And Nitric Acids Part I: Calculation Of Corrosion Depth. *Cem. Concr. Res.* **1994**, *24*, 551−562.
(23) Giraudo, N.; Thissen, P. Carbonation Competing Functionalization On Calcium-Silicate-Hydrates: Investigation Of Four Promising Surface-Activation Techniques. *ACS Sustainable Chem. Eng.* **2016**, *4*, 3985−3994.
(24) Šavija, B.; Luković, M. Carbonation Of Cement Paste: Understanding, Challenges, And Opportunities. *Constr. Build. Mater.* **2016**, *117*, 285−301.
(25) Haselbach, L. M.; Ma, S. Potential For Carbon Adsorption On Concrete: Surface XPS Analyses. *Environ. Sci. Technol.* **2008**, *42*, 5329−5334.
(26) Zhang, D.; Ghouleh, Z.; Shao, Y. Review On Carbonation Curing Of Cement-Based Materials. *J. CO2 Util.* **2017**, *21*, 119−131.
(27) Ghouleh, Z.; Guthrie, R. I. L.; Shao, Y. Production Of Carbonate Aggregates Using Steel Slag And Carbon Dioxide For Carbon-Negative Concrete. *J. CO2 Util.* **2017**, *18*, 125−138.
(28) Han, J.; Liang, Y.; Sun, W.; Liu, W.; Wang, S. Microstructure Modification Of Carbonated Cement Paste With Six Kinds Of Modern Microscopic Instruments. *J. Mater. Civ. Eng.* **2015**, *27*, No. 04014262.
(29) Morandeau, A.; Thiéry, M.; Dangla, P. Investigation Of The Carbonation Mechanism Of CH And C-S-H In Terms Of Kinetics, Microstructure Changes And Moisture Properties. *Cem. Concr. Res.* **2014**, *56*, 153−170.
(30) Young, J. F.; Berger, R. L.; Breese, J. Accelerated Curing Of Compacted Calcium Silicate Mortars On Exposure To CO2. *J. Am. Ceram. Soc.* **1974**, *57*, 394−397.
(31) Lu, L.; Ouyang, D. Properties Of Cement Mortar And Ultra-High Strength Concrete Incorporating Graphene Oxide Nanosheets. *Nanomaterials* **2017**, *7*, 187.
(32) Lv, S.; Ma, Y.; Qiu, C.; Sun, T.; Liu, J.; Zhou, Q. Effect Of Graphene Oxide Nanosheets Of Microstructure And Mechanical Properties Of Cement Composites. *Constr. Build. Mater.* **2013**, *49*, 121−127.
(33) Pan, Z.; He, L.; Qiu, L.; Korayem, A. H.; Li, G.; Zhu, J. W.; Collins, F.; Li, D.; Duan, W. H.; Wang, M. C. Mechanical Properties And Microstructure Of A Graphene Oxide−Cement Composite. *Cem. Concr. Compos.* **2015**, *58*, 140−147.
(34) Mohammed, A.; Sanjayan, J. G.; Nazari, A.; al-Saadi, N. T. K. The Role Of Graphene Oxide In Limited Long-Term Carbonation Of Cement-Based Matrix. *Constr. Build. Mater.* **2018**, *168*, 858−866.
(35) Li, X.; Lu, Z.; Chuah, S.; Li, W.; Liu, Y.; Duan, W. H.; Li, Z. Effects Of Graphene Oxide Aggregates On Hydration Degree, Sorptivity, And Tensile Splitting Strength Of Cement Paste. *Composites, Part A* **2017**, *100*, 1−8.
(36) Lu, Z.; Hou, D.; Meng, L.; Sun, G.; Lu, C.; Li, Z. Mechanism Of Cement Paste Reinforced By Graphene Oxide/Carbon Nanotubes Composites With Enhanced Mechanical Properties. *RSC Adv.* **2015**, *5*, 100598−100605.
(37) Lv, S.; Hu, H.; Zhang, J.; Luo, X.; Lei, Y.; Sun, L. Fabrication Of GO/Cement Composites By Incorporation Of Few-Layered GO Nanosheets And Characterization Of Their Crystal/Chemical Structure And Properties. *Nanomaterials* **2017**, *7*, 457.
(38) Yaseen, S. A.; Yiseen, G. A.; Li, Z. Synthesis Of Calcium Carbonate In Alkali Solution Based On Graphene Oxide And Reduced Graphene Oxide. *J. Solid State Chem.* **2018**, *262*, 127−134.
(39) Serp, P.; Machado, B. *Nanostructured Carbon Materials For Catalysis*; Royal Society of Chemistry: Cambridge, 2015.
(40) Paredes, J. I.; Villar-Rodil, S.; Martínez-Alonso, A.; Tascón, J. M. D. Graphene Oxide Dispersions In Organic Solvents. *Langmuir* **2008**, *24*, 10560−10564.
(41) Boehm, H.-P.; Diehl, E.; Heck, W.; Sappok, R. Surface Oxides Of Carbon. *Angew. Chem., Int. Ed. Engl.* **1964**, *3*, 669−677.
(42) Boehm, H. P. Chemical identification of surface groups. In: *Advances In Catalysis*. Eley, D.; Pines, H.; Weisz, P.; eds., 1st ed.; Academic Press: New York, 1966; pp. 179−274.
(43) Boehm, H. P. Some Aspects Of The Surface Chemistry Of Carbon Blacks And Other Carbons. *Carbon* **1994**, *32*, 759−769.
(44) Boehm, H. P. Surface chemical characterization of carbons from adsorption studies. In *Adsorption By Carbons*. Bottani, E.; Tascón, J.; eds., 1st ed.; Elsevier Science: Amsterdam, 2008; pp. 301−327.
(45) Boehm, H. P. Carbon surface chemistry. *Graphite And Precursors*; 1st ed.; Gordon and Breach Science Publishers: Amsterdam, 2001; pp. 141−179.
(46) Goertzen, S. L.; Thériault, K. D.; Oickle, A. M.; Tarasuk, A. C.; Andreas, H. A. Standardization Of The Boehm Titration. Part I. CO 2 Expulsion And Endpoint Determination. *Carbon* **2010**, *48*, 1252−1261.
(47) Pratt, P. L.; Jennings, H. M. The Microchemistry And Microstructure Of Portland Cement. *Annu. Rev. Mater. Sci.* **1981**, *11*, 123−149.
(48) Stutzman, P. E.; Feng, P.; Bullard, J. W. Phase Analysis Of Portland Cement By Combined Quantitative X-Ray Powder Diffraction And Scanning Electron Microscopy. *J. Res. Natl. Inst. Stand. Technol.* **2016**, *121*, 47−107.
(49) Mollah, M. Y. A.; Yu, W.; Schennach, R.; Cocke, D. L. A Fourier Transform Infrared Spectroscopic Investigation Of The Early Hydration Of Portland Cement And The Influence Of Sodium Lignosulfonate. *Cem. Concr. Res.* **2000**, *30*, 267−273.
(50) Ylmén, R.; Jäglid, U.; Steenari, B.-M.; Panas, I. Early Hydration And Setting Of Portland Cement Monitored By IR, SEM And Vicat Techniques. *Cem. Concr. Res.* **2009**, *39*, 433−439.
(51) Jumate, E.; Manea, D. X-Ray Diffraction Study of Hydration Processes in the Portland Cement. *J. Eng. Appl. Sci.* **2011**, *14*, 79−86.
(52) Stepkowska, E. T.; Pérez-Rodríguez, J. L.; Sayagués, M. J.; Martínez-Blanes, J. M. Calcite, Vaterite and Aragonite Forming on Cement Hydration from Liquid and Gaseous Phase. *J. Therm. Anal. Calorim.* **2003**, *73*, 247−269.
(53) Peter, M. A.; Muntean, A.; Meier, S. A.; Böhm, M. Competition Of Several Carbonation Reactions In Concrete: A Parametric Study. *Cem. Concr. Res.* **2008**, *38*, 1385−1393.
(54) Cizer, Ö.; Van Balen, K.; Elsen, J.; Van Gemert, D. Real-Time Investigation Of Reaction Rate And Mineral Phase Modifications Of Lime Carbonation. *Constr. Build. Mater.* **2012**, *35*, 741−751.
(55) Martínez-Ramírez, S.; Fernández-Carrasco, L. Carbonation Of Ternary Cement Systems. *Constr. Build. Mater.* **2012**, *27*, 313−318.
(56) Grounds, T.; G Midgley, H.; V Novell, D. Carbonation Of Ettringite By Atmospheric Carbon Dioxide. *Thermochim. Acta* **1988**, *135*, 347−352.

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DOI: 10.1021/acsomega.9b00042
ACS Omega 2019, 4, 10160−10170

(57) Zhou, Q.; Glasser, F. P. Kinetics And Mechanism Of The Carbonation Of Ettringite. *Advances in Cement Research* **2000**, *12*, 131−136.
(58) Xiantuo, C.; Ruizhen, Z.; Xiaorong, C. Kinetic Study Of Ettringite Carbonation Reaction. *Cem. Concr. Res.* **1994**, *24*, 1383−1389.
(59) Hyvert, N.; Sellier, A.; Duprat, F.; Rougeau, P.; Francisco, P. Dependency Of C−S−H Carbonation Rate On CO2 Pressure To Explain Transition From Accelerated Tests To Natural Carbonation. *Cem. Concr. Res.* **2010**, *40*, 1582−1589.
(60) Yousuf, M.; Mollah, A.; Hess, T. R.; Tsai, Y.-N.; Cocke, D. L. An FTIR And XPS Investigations Of The Effects Of Carbonation On The Solidification/Stabilization Of Cement Based Systems-Portland Type V With Zinc. *Cem. Concr. Res.* **1993**, *23*, 773−784.
(61) Horgnies, M.; Chen, J. J.; Bouillon, C. Overview About The Use Of Fourier Transform Infrared Spectroscopy To Study Cementitious Materials. *Materials Characterisation VI* **2013**, *77*, 251−262.
(62) Xyla, A. G.; Koutsoukos, P. G. Quantitative Analysis Of Calcium Carbonate Polymorphs By Infrared Spectroscopy. *J. Chem. Soc., Faraday Trans. 1* **1989**, *85*, 3165−3172.
(63) Zhang, L.; Glasser, F. P. Investigation of the microstructure and carbonation of CS^{-}A-based concretes removed from service. *Cem. Concr. Res.* **2005**, *35*, 2252−260.
(64) Gunasekaran, S.; Anbalagan, G.; Pandi, S. Raman And Infrared Spectra Of Carbonates Of Calcite Structure. *J. Raman Spectrosc.* **2006**, *37*, 892−899.
(65) Kalijadis, A. M.; Vukcevic, M. M.; Jovanovic, Z. M.; Lausevic, Z. V.; Lausevic, M. D. Characterization Of Surface Oxygen Groups On Different Carbon Materials By The Boehm Method And Temperature Programmed Desorption. *J. Serb. Chem. Soc.* **2011**, *76*, 757−768.

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DOI: 10.1021/acsomega.9b00042
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